

Binary Classifiers as Dilations

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Microeconomic Colloquium



BERLIN SCHOOL
OF ECONOMICS

First a bit about me

- ▷ Currently, a Lecturer of Economics at the University of Edinburgh
 - ▷ Visiting Assistant Professor at University of Pittsburgh, 2022–23

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- ▷ Research Interests (so far):
 - ▷ Game Theory (so far, mostly epistemic game theory and robustness concerns)
 - ▷ (Applied) Decision Theory
 - ▷ School Choice (Matching)

Binary Diagnostic Tests



- ▷ Medical diagnostic tests:
 - ▷ HIV test, caries, cancer screening, diabetes, pulmonary embolism...
 - ▷ Antigen tests: Covid, influenza, RSV, streptococcus, ...
 - ▷ Antibody/ELISA tests

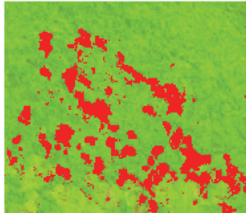
Binary Diagnostic Tests



- ▷ Medical diagnostic tests:
 - ▷ HIV test, caries, cancer screening, diabetes, pulmonary embolism...
 - ▷ Antigen tests: Covid, influenza, RSV, streptococcus, ...
 - ▷ Antibody/ELISA tests
- ▷ Binary expert testing, binary classification, ...

Land Cover Change

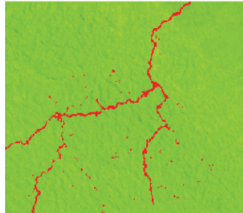
Small-scale agriculture



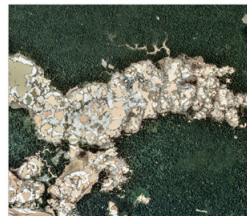
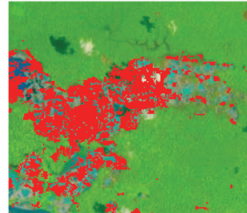
Large-scale agriculture



Logging roads



Gold mining



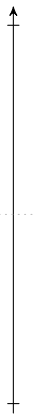
Finer, Matt, et al. "Combating deforestation: From satellite to intervention." *Science* (2018).

Usual Binary Classifiers

$P(\text{inf.})$ before test



$P(\text{inf.})$ after test



▷ Stick to medical interpretation:

Usual Binary Classifiers

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Positive test result



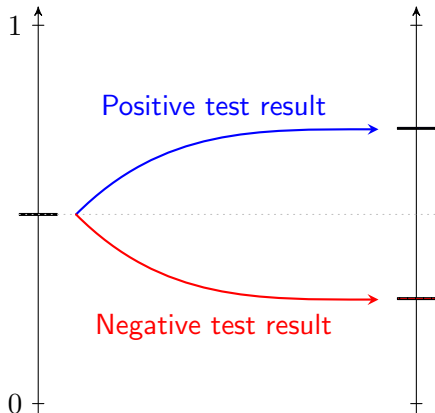
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Usual Binary Classifiers

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Usual Binary Classifiers

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Positive test result

Negative test result

$P(\text{inf.})$ after test



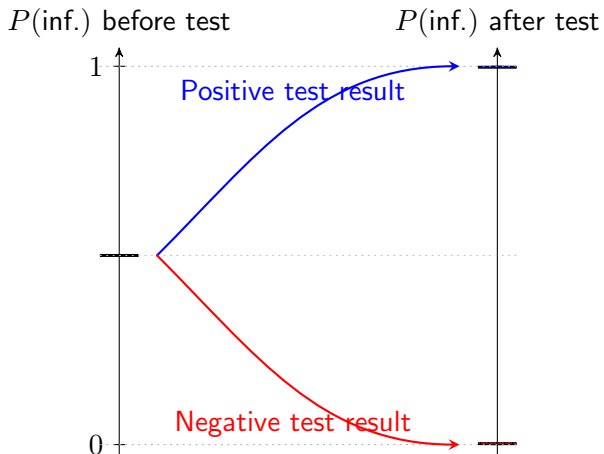
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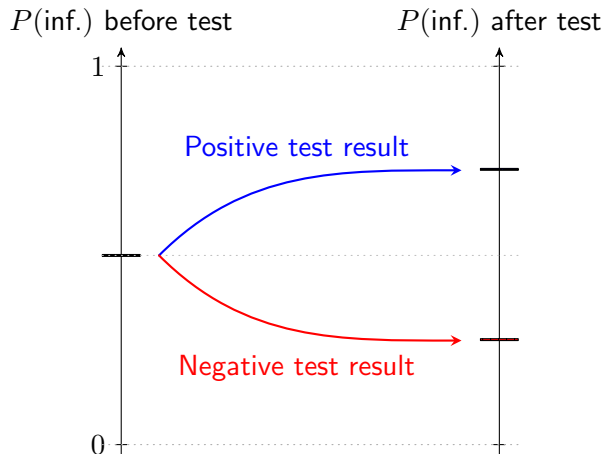
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Usual Binary Classifiers



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Usual Binary Classifiers



- ▷ Stick to medical interpretation:
 - ▷ Positive result, or
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- ▷ Possible extremes:
 - ▷ No information
 - ▷ Full information
- ▷ always (weakly) informative à la Blackwell

Tests with ambiguous information

$P(\text{inf.})$ before test



$P(\text{inf.})$ after test



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Positive test result

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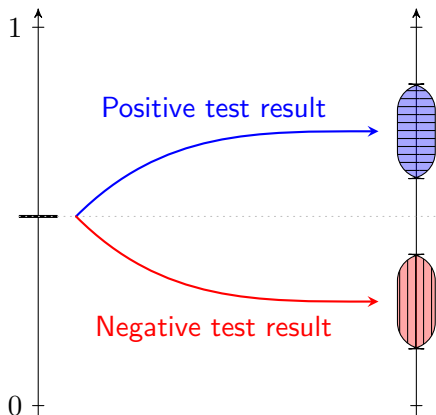


After test result: set of posteriors
aka *ambiguity*/*Knightian uncertainty*

Tests with ambiguous information

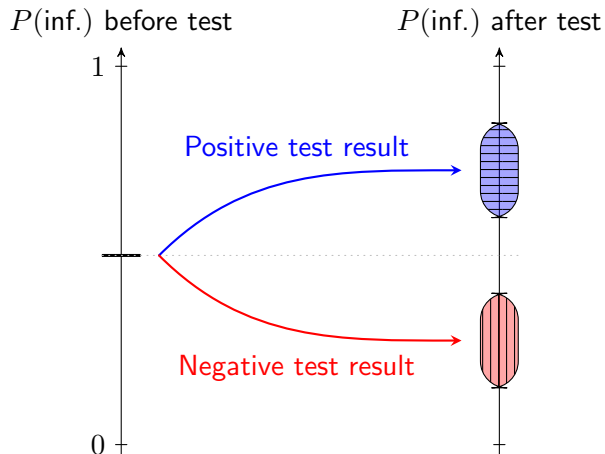
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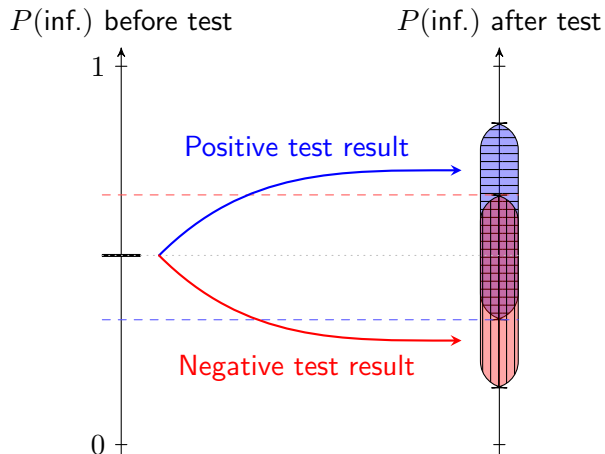


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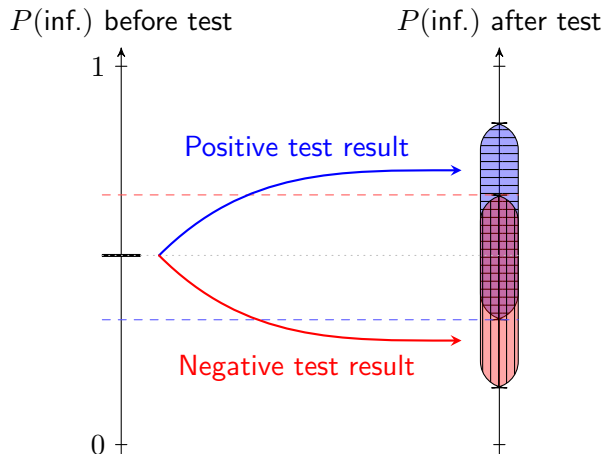


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- ▷ Here: **Robust approach** to allow for all possible correlations
 - ▷ Possible interpretation: Ambiguity or group of disagreeing experts

Evaluating new tests

Policy maker might want know when new test is a dilation

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4. Empirical application: Covid LFT tests , but other procedures , ML prediction

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 - Bose and Renou (2014), Beauchêne, Li, and Li (2019), Cheng (2021), Pahlke (2022)
 - ▷ Recently, also experimental evidence: in particular, Shishkin and Ortoleva (2023)
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- ▷ This paper is the first illustration of this curious property “in the field”
 - ▷ But see Manski (2018)

Outline

1. Model
2. Illustrations & Applications
3. Statistical Inference (work in progress)
4. Extension and Conclusions

1 – Model

New test: Accuracy

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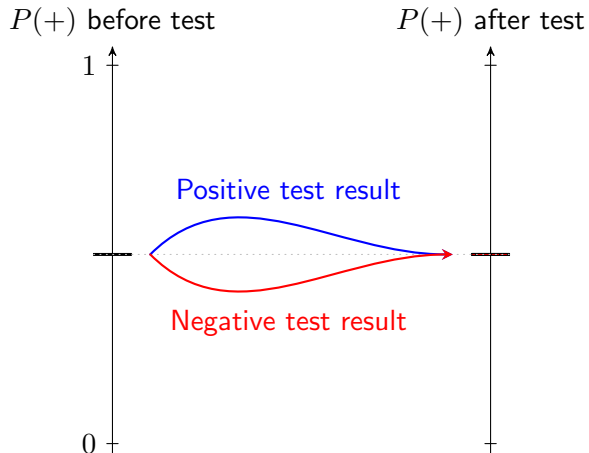
$$\mathbb{P}(t = 1|+) =: \theta_1 \quad (\text{Sensitivity})$$

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Assumption (Non-triviality)

The population satisfies $\mathbb{P}(+) \in (0, 1)$.

No Information

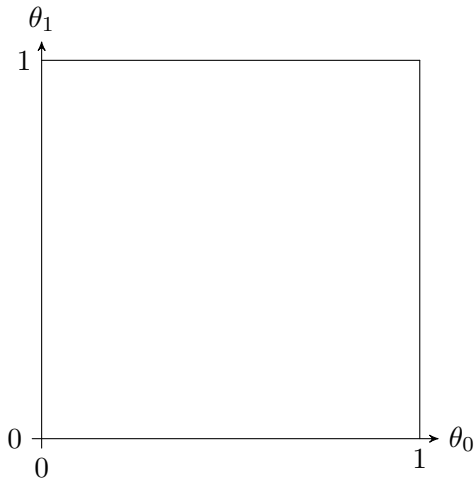


Observation

The new test provides no information (in Blackwell sense) if and only if

$$\theta_0 + \theta_1 = 1.$$

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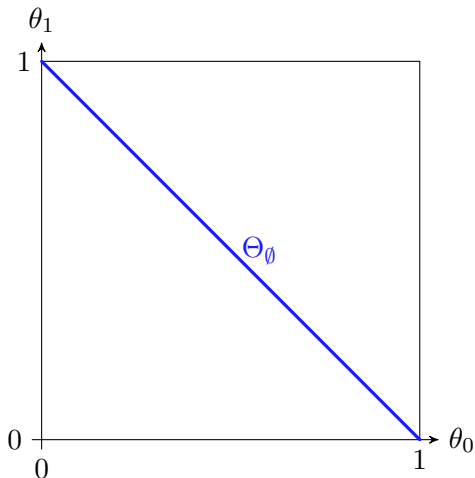


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$$\Theta_{\emptyset} := \{(\theta_0, \theta_1) : \theta_0 + \theta_1 = 1\}$$

Gold Standards

To obtain $\theta = (\theta_0, \theta_1)$ true infection needs to be observed

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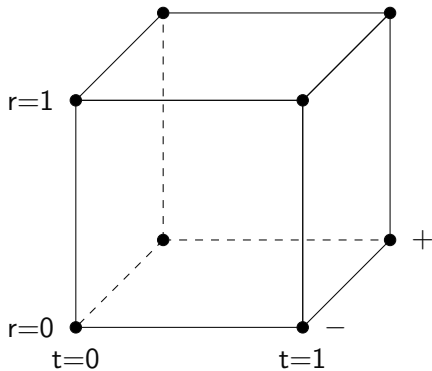
Data: $\mathbb{P}(r, t)$

Need: $\mathbb{P}(t|\pm)$ (or $\mathbb{P}(r, t, \pm)$)

Multiplicity and Ambiguous Information

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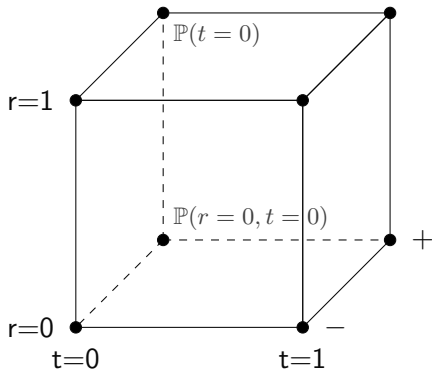
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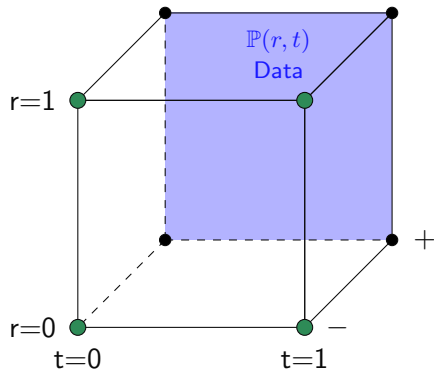
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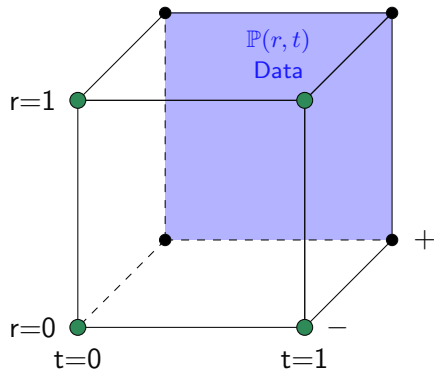
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The emergence of ambiguity:

- ▷ Multiple $\mathbb{P}(r, t, \pm) \in: \Pi$



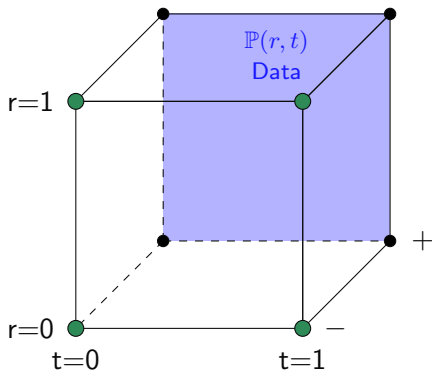
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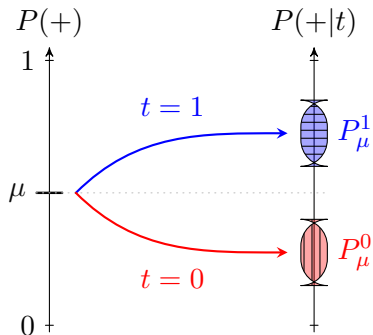
- ▷ Multiple $\mathbb{P}(r, t, \pm) \in: \Pi$
- ▷ Multiple $\theta \in: \Theta$ possible
- ▷ One *ambiguous* experiment or
a *set* of Blackwell experiments



Dilating test

For $\mu \in \Delta(\pm)$, define

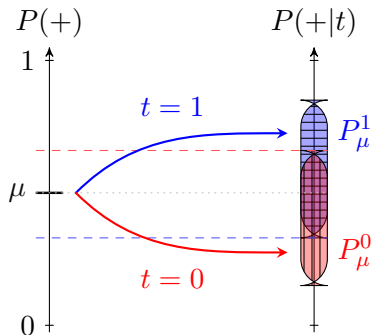
$$P_\mu^1 := \left\{ \frac{\mu\theta_1}{\mu\theta_1 + (1-\mu)(1-\theta_0)} : \theta \in \Theta \right\} \text{ and } P_\mu^0 := \left\{ \frac{\mu(1-\theta_1)}{\mu(1-\theta_1) + (1-\mu)\theta_0} : \theta \in \Theta \right\}$$



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Definition

Fix a prior $\mu \in \Delta(\pm)$. The new test is a

μ -dilation if $\{\mu\} \subsetneq P_\mu^0 \cap P_\mu^1$.

It's a *dilation* if it's a μ -dilation for all $\mu \in \Delta(\pm)$.

Reference for what?

Observation

For any data $\mathbb{P}(r, t)$ and assuming non-triviality, the new test is a dilation.

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Assumption (Knowledge of the reference test)

Reference test's sensitivity $\mathbb{P}(r = 1|+) =: s_1$ and specificity $\mathbb{P}(r = 0|-) =: s_0$ are known.

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Assumption (Informative reference test)

The reference test satisfies $s_1 + s_0 > 1$.

Characterization of the ambiguous experiment

Proposition

Fix data $\mathbb{P}(t, r)$. Under the assumptions the following hold:

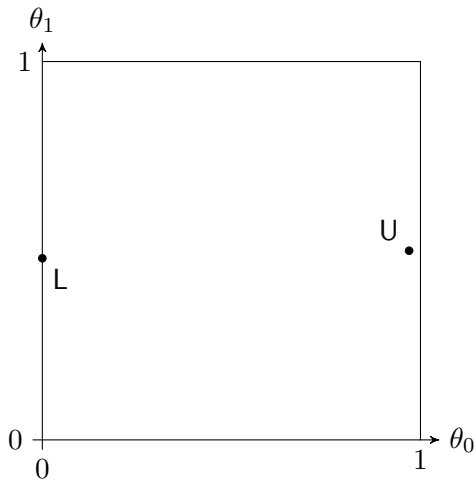
$$\Theta(s) := \left\{ (\theta_0, \theta_1) : \theta_1 \in [L_s, U_s], \mathbb{P}_s(-)\theta_0 - \mathbb{P}_s(+)\theta_1 = \mathbb{P}(t = 0) \right\}.$$

$$\mathbb{P}_s(+) = \frac{\mathbb{P}(r = 1) + s_0 - 1}{s_1 + s_0 - 1} \quad \mathbb{P}_s(-) = 1 - \mathbb{P}_s(+)$$

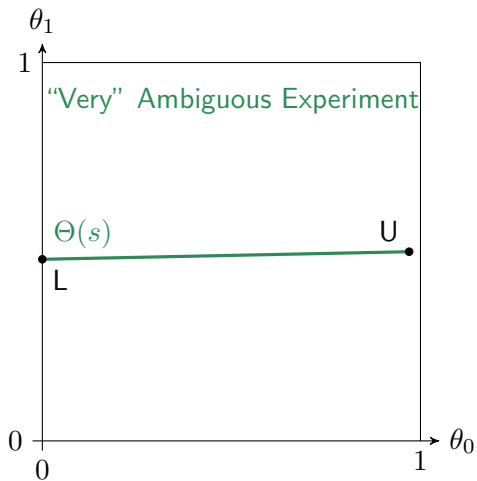
$$L_s = \frac{[\mathbb{P}(t = 1, r = 0) - s_0 \mathbb{P}_s(-)]^+ + [\mathbb{P}(t = 1, r = 1) - (1 - s_0) \mathbb{P}_s(-)]^+}{\mathbb{P}_s(+)}$$

$$U_s = \frac{\min\{\mathbb{P}(t = 1, r = 0), (1 - s_1) \mathbb{P}_s(+)\} + \min\{\mathbb{P}(t = 1, r = 1), s_1 \mathbb{P}_s(+)\}}{\mathbb{P}_s(+)}$$

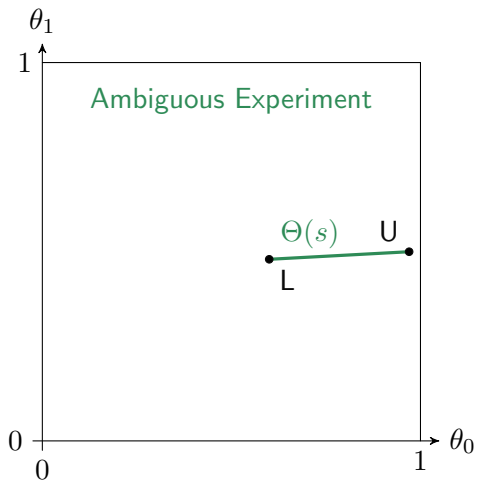
Visualizing the characterization



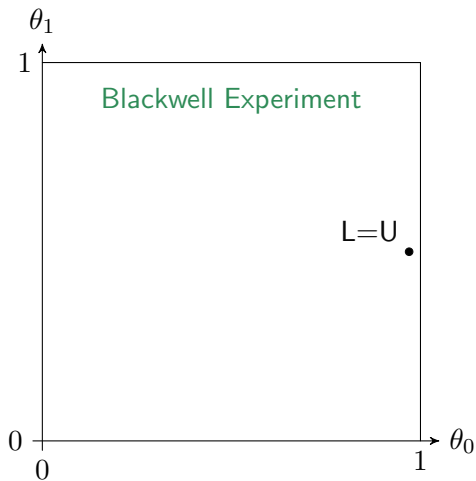
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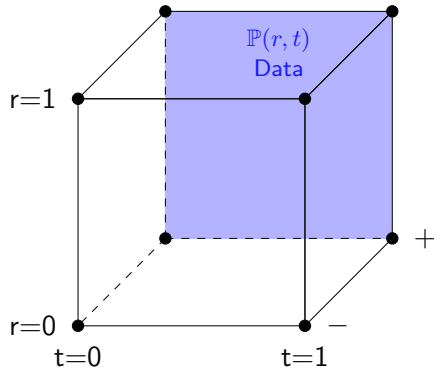


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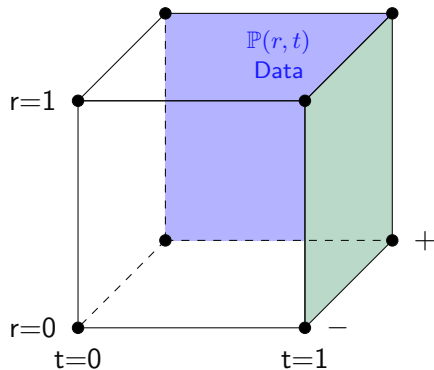
1 – Model

Proof



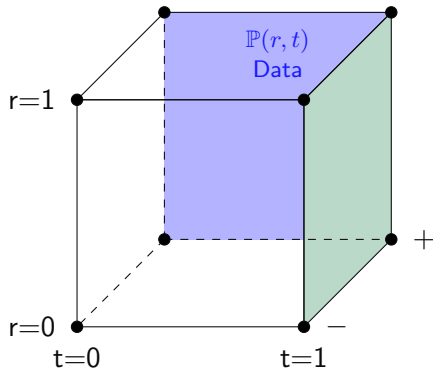
Proof

1. Informative reference $\implies \mathbb{P}(r, \pm)$



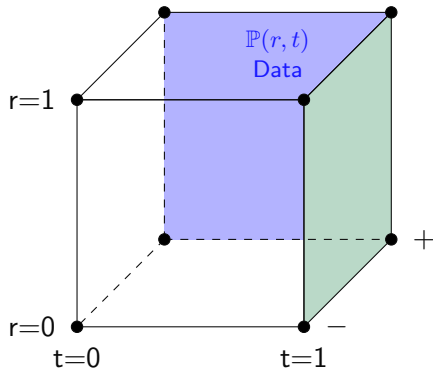
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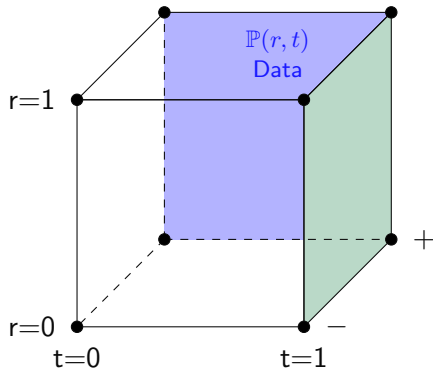
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4. U/L from most/least correlated distributions



The need for ambiguity

Recall: Definition of dilation required $\{\mu\} \subsetneq P_\mu^0 \cap P_\mu^1$

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A Blackwell experiment (i.e. $|\Theta| = 1$) cannot be a dilation.

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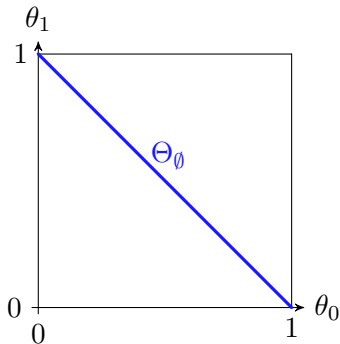
Lemma 4: Together with other assumptions, sufficient for proper ambiguity, i.e. $|\Theta| > 1$

Characterizing Dilations

Theorem

Under the mentioned assumptions,

t is a dilation if and only if there exists $\theta \in \Theta$ such that $\theta_1 + \theta_0 = 1$.

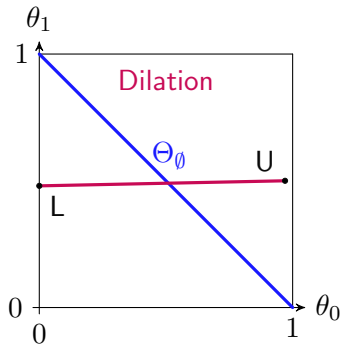


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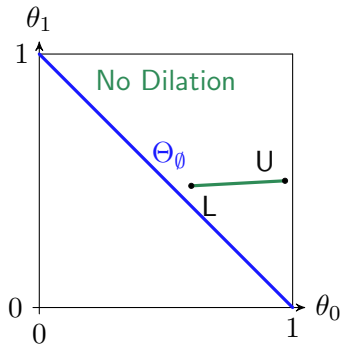


Characterizing Dilations

Theorem

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Corollary: Dilation if and only if $\mathbb{P}(t = 1) = \theta_1$ for some $\theta \in \Theta$.

That is, there is at most one intersection point with $\Theta_\emptyset$.

2 – Illustrative Applications

Application 1: Covid Antigen test (LFT) from Leber et al. (2021)

new \ ref.	$r = 0$	$r = 1$	Sum
$t = 0$	179	38	217
$t = 1$	22	788	819
Sum	201	826	1027

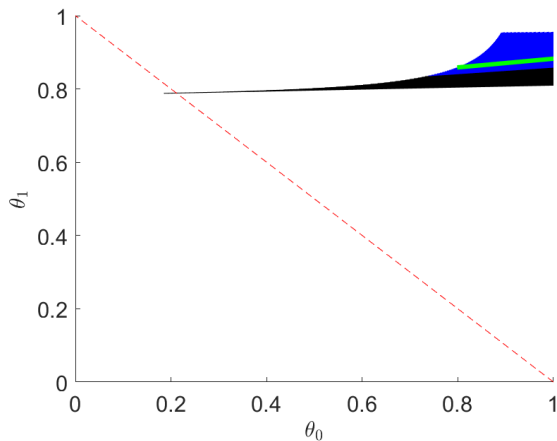
- ▷ “Patients with ... flu-like symptoms attending a general practice network in an Austrian district” – Leber et al. (2021, p.1)
- ▷ Apparent Sensitivity: $\mathbb{P}(t = 1|r = 1) = 95.4\%$
- ▷ Apparent Specificity: $\mathbb{P}(t = 0|r = 0) = 89.1\%$

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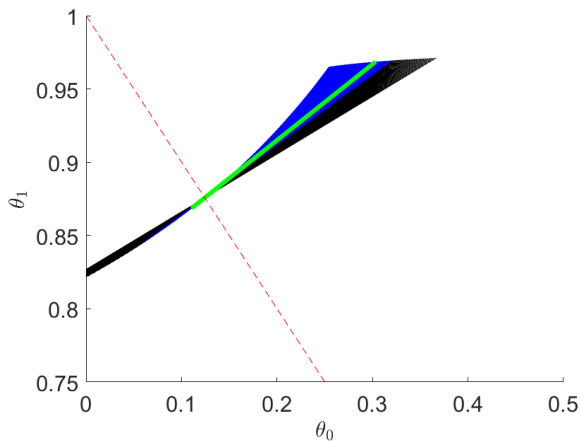
- ▷ Apparent Sensitivity: $\mathbb{P}(t = 1|r = 1) = 96.5\%$
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3 – Statistical Inference

Statistical Test(s) for Dilations

Hypothesis: $H_0 =$ new test is a dilation ($\iff \exists(\theta_1, \theta_0) \in \Theta : \theta_1 + \theta_0 = 1$)

First, somewhat naive, attempt based on Goodman (1965)

new \ ref.	$r = 0$	$r = 1$
$t = 0$	p_{00}	p_{01}
$t = 1$	p_{10}	p_{11}

- ▷ Data $\sim$ multinomial distribution
- ▷ Goodman (1965): confidence set for parameters
- ▷ Get CI for $\theta_0 + \theta_1$, reject if 1 not in it

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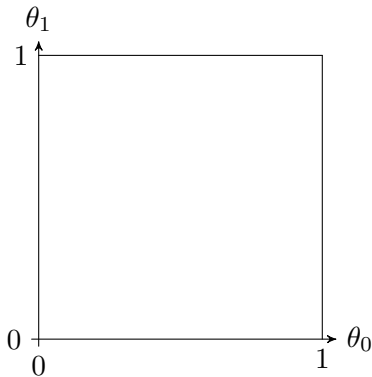
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- ▷ Get CI for $\theta_0 + \theta_1$, reject if 1 not in it
- ▷ Issues:
 - ▷ Conservative
 - ▷ Low power
 - ▷ Uniformity problems

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Key insight from Obradović (2024): Θ can be expressed via **moment inequalities**



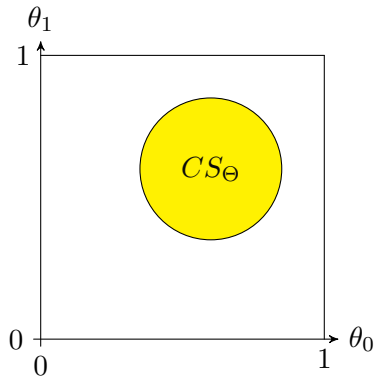
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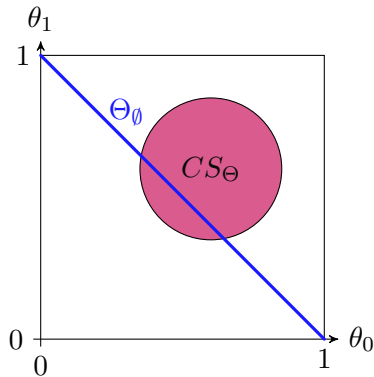
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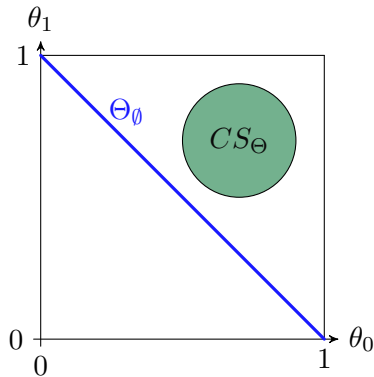
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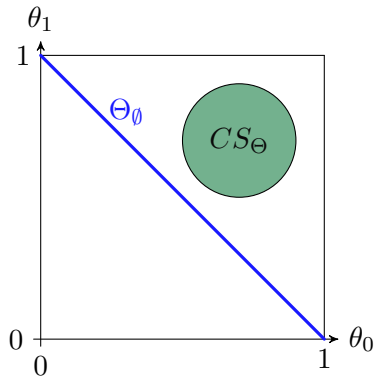
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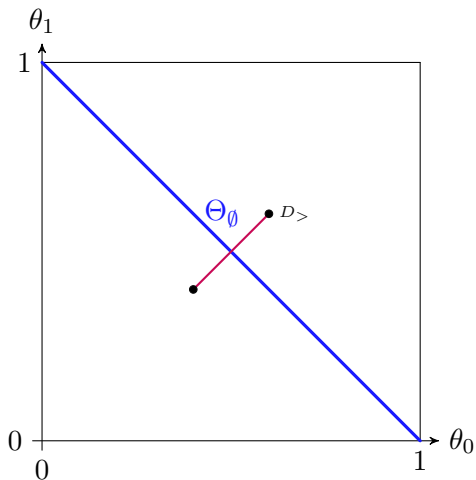
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- ▷ ... but only need to look across $\Theta_\emptyset$

2. **Subvector inference:**

- ▷ based on Bugni, Canay, and Shi (2017)
- ▷ better coverage and power

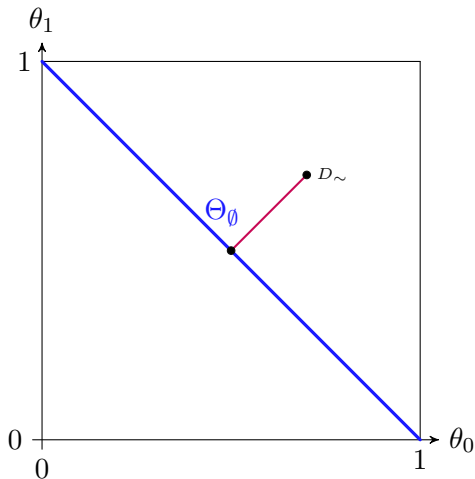
Simulations



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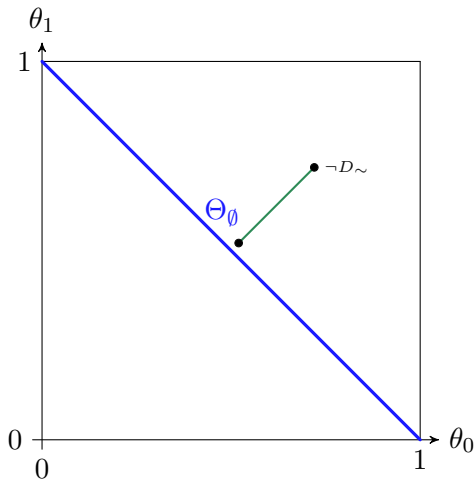
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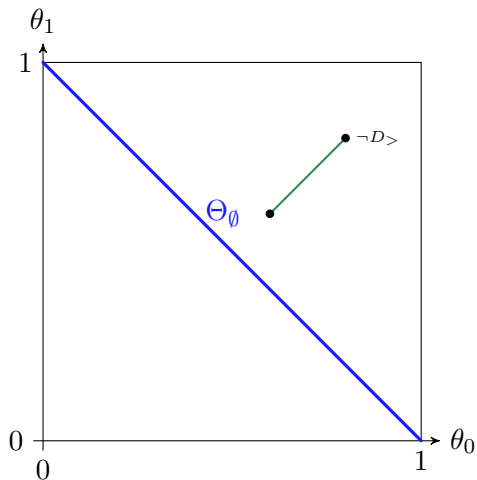
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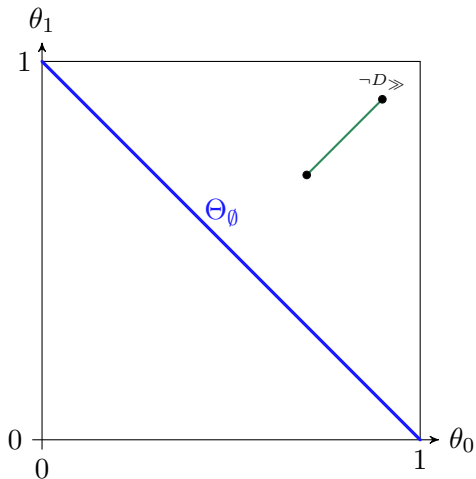
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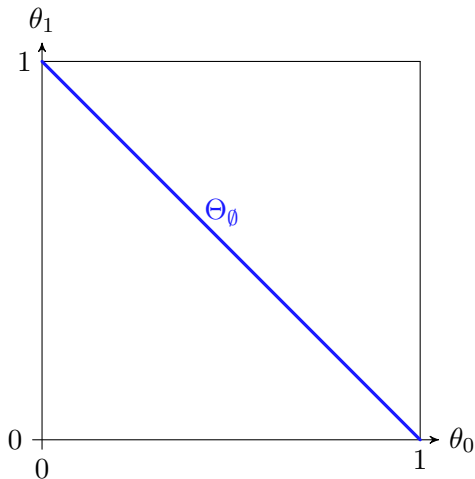


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- ▷ Sample Size:
 - ▷ $N \in \{50, 100, 500\}$
- ▷ Simulations replications: 1000

Simulations - Rejection Probabilities ($\alpha = 5\%$)

		$D_{>}$	$D_{\sim}$	$\neg D_{\sim}$	$\neg D_{>}$	$\neg D_{\gg}$
$N = 50$	G	0.9%	0.6%			
	RSW	0%	0%			
	BCS	0%	0.8%			
$N = 100$	G	0%	0%			
	RSW	0%	0%			
	BCS	0%	1%			
$N = 500$	G	0%	0%			
	RSW	0%	0%			
	BCS	0%	1.3%			

Simulations - Rejection Probabilities ($\alpha = 5\%$)

		$D_{>}$	$D_{\sim}$	$\neg D_{\sim}$	$\neg D_{>}$	$\neg D_{\gg}$
$N = 50$	G	0.9%	0.6%	0.2%	0.7%	4.3%
	RSW	0%	0%	0%	0.1%	4.3%
	BCS	0%	0.8%	3.9%	22%	76%
$N = 100$	G	0%	0%	0%	1%	30%
	RSW	0%	0%	0%	0.1%	19%
	BCS	0%	1%	3.1%	43%	100%
$N = 500$	G	0%	0%	0%	50%	100%
	RSW	0%	0%	0%	28%	100%
	BCS	0%	1.3%	9.2%	99%	100%

Moment (in)equalities

Fix $P = \mathbb{P}(r, t)$. Obradović (2024) shows that $\Theta(s)$ is can be rewritten as

$$\tilde{\Theta}_P(s) = \left\{ \theta \in [0, 1]^2 : \mathbb{E}_P[m_j(r, t; \theta, s)] \leq 0, j = 1, \dots, 6 \text{ and } \mathbb{E}_P[m_j(r, t; \theta, s)] = 0 \right\},$$

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▷ **Moment Inequality Model**

We are interested in testing the hypothesis that $H_0: \theta_1 + \theta_0 = 1$, i.e. based on a linear function of the parameters.

▷ **Subvector Inference**

▷ Bugni, Canay, and Shi (2017) propose bootstrap-based inference method

Assumptions

We consider a (relatively large) class of distributions $\mathcal{P} \subset \Delta^3 \subset \mathbb{R}^4$.

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Then, strengthen identification assumptions to avoid (well-known) uniformity problems:

- ▷ Without: For any n , there might exist $P \in \mathcal{P}$ rendering confidence regions invalid.
- ▷ Often better finite sample performance if uniformity holds.
- ▷ Especially important for partially identified models.

Stronger Assumptions

Want the probability of infection to be strictly (and uniformly) bounded away from boundary:

Assumption

There exists $\varepsilon_y \in (0, \frac{1}{2})$ such that for all $P \in \mathcal{P}$, we have $\mathbb{P}_s(+) \in [\varepsilon_y, 1 - \varepsilon_y]$.

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Strengthen “Everything goes” assumption similarly:

Assumption

There exists an $\varepsilon_d \in (0, \frac{1}{4})$ s.t. for every $P \in \mathcal{P}$ and every $(t, r) \in \{0, 1\}^2$, $P(t, r) \geq \varepsilon_d$ holds.

Asymptotic Size Control

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$$Q_P(\theta, s) = \sum_{j=1}^6 \left[\max \left\{ 0, \frac{\mathbb{E}_P m_j(W, \theta, s)}{\sigma_{P,j}(\theta, s)} \right\} \right]^2 + \left[\frac{\mathbb{E}_P m_7(W, \theta, s)}{\sigma_{P,j}(\theta, s)} \right]^2$$

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- ▷ We show Bugni et al (2017, Thm 4.1) applies, so our test asympt. and unif. controls size:

$$\limsup_{n \rightarrow \infty} \sup_{(\theta, P): H_0 \text{ is true}} \mathbb{P}(\text{Type 1 error}) \leq \alpha.$$

4 – Applications

CT-Chest scan for Covid from Ai et al. (2020)

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Test statistic $T_n = 1.2510^{-18}$

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Critical value $\hat{c}_n^{95\%} = 1.11$

Cannot reject H_0 of it being a dilation
(p -value $> 99\%$)

Deep Neural Net to Predict Loan Approval (Abakarim et al. '18)

- ▷ Predict “good” ($t = 1$) vs. “bad” risk ($t = 0$)

Deep Neural Net to Predict Loan Approval (Abakarim et al. '18)

new \ ref.	$r = 0$	$r = 1$	Sum
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 Reject H_0 : $T_n = 35.96$ vs. $\hat{c}_n^{95\%} = 5.25$

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 - and historic data $\neq$ ground truth
- ▷ Assume $\alpha = 5\%$ and $s = (0.9, 0.9)$
 - Reject H_0 : $T_n = 35.96$ vs. $\hat{c}_n^{95\%} = 5.25$
 - Also reject with $s = (0.95, 0.95)$

5 – Extensions and Conclusion

Extensions: Partial Knowledge about Reference Test

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 - ▷ Moment inequality/subvector inference too
 - ▷ However, we were not able (yet) to prove size control

Extensions: No Knowledge about Reference Test

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- ▷ Given data $\rightarrow$ find $s = (s_0, s_1)$ for dilation
- ▷ Thinking of $\Theta(s)$ as correspondence: $\mathcal{D} := \Theta^{-1}(\Theta_\emptyset)$
- ▷ Again, a moment inequality approach is applicable

Illustration 1: Independent tests

new \ ref.	$r = 0$	$r = 1$	Sum
$t = 0$	25%	25%	50%
$t = 1$	25%	25%	50%
Sum	50%	50%	

Illustration 1: Independent tests

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$t = 1$	25%	25%	50%
Sum	50%	50%	

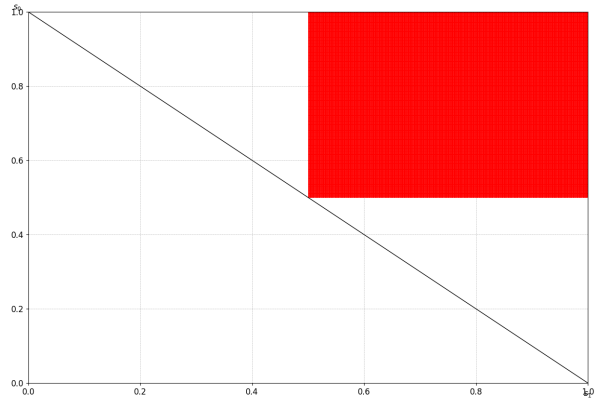


Illustration 2: Weakly correlated tests

new \ ref.	$r = 0$	$r = 1$	Sum
$t = 0$	39%	11%	50%
$t = 1$	11%	39%	50%
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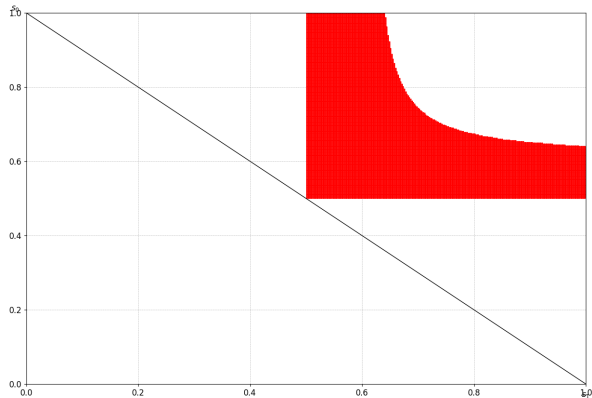
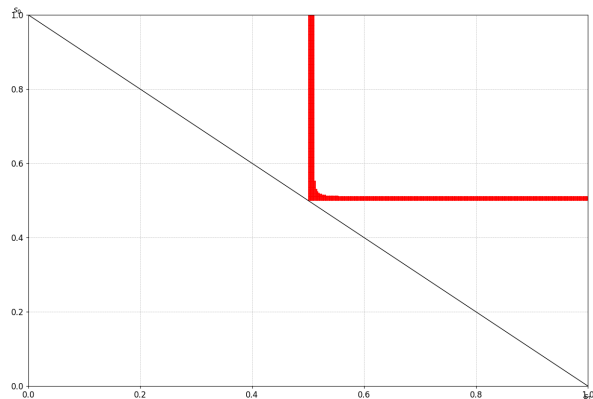


Illustration 3: Highly correlated tests

new \ ref.	$r = 0$	$r = 1$	Sum
$t = 0$	49%	1%	50%
$t = 1$	1%	49%	50%
Sum	50%	50%	

Illustration 3: Highly correlated tests

new \ ref.	$r = 0$	$r = 1$	Sum
$t = 0$	49%	1%	50%
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Policy implication: Regulation of medical tests

new \ ref.	$r = 0$	$r = 1$	Sum
$t = 0$	388	120	508
$t = 1$	12	480	492
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▷ If reference test is perfect:

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 - ▷ 97% specificity and
 - ▷ 80% sensitivity

- ▷ **BUT:** even if perfect specificity of reference
if s_1 close to 62%
we have a dilation

Further research

- ▷ Partial knowledge of correlation, *e.g.* local robustness around independence

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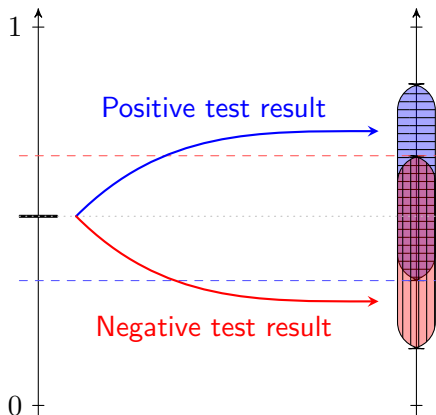
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- ▷ Theory of ambiguous information

Summary

Do we encounter dilations in (existing?) classifiers?



Thanks! – Gabriel Ziegler, U Edinburgh, ziegler@ed.ac.uk